

## YELLOWSTONE, WY, USA

WMO: 726664

Lat: 44.544N

Lon: 110.421W

Elev: 7835

StdP: 10.98

Time Zone: -7.00 (NAM)

Period: 94-19

WBAN: 94173

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |       | Humidification DP/MCDB and HR |     |       |       |     |       | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-------|-------------------------------|-----|-------|-------|-----|-------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |       | 99.6%                         |     |       | 99%   |     |       | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%   | DP                            | HR  | MCDB  | DP    | HR  | MCDB  | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)   | (d)                           | (e) | (f)   | (g)   | (h) | (i)   | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | -19.3      | -12.4 | -23.9                         | 2.0 | -18.0 | -17.4 | 2.8 | -11.2 | 12.9                  | 22.3 | 11.7 | 23.5 | 1.0                   | 310  | 0.699 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 31.5                   | 77.2            | 52.0 | 74.4 | 51.4 | 71.7 | 50.7 | 56.1                | 68.5 | 54.5 | 67.4 | 53.0 | 66.3 | 6.0                  | 270  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 51.6                            | 76.3 | 59.3 | 49.4 | 70.1 | 58.6 | 47.4 | 65.0 | 57.1 | 27.0          | 68.1 | 25.8 | 67.0 | 24.8 | 66.3 | 62.8              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|
| 1%                | 2.5% | 5%  | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |
|                   |      |     | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |
| 12.4              | 10.7 | 9.3 | -31.0                      | 81.0 | 3.4                | 2.1 | -33.4                                              | 82.5 | -35.5      | 83.7 | -37.4      | 84.8 | -39.9      | 86.3 |
|                   |      |     | -29.2                      | 59.7 | 2.5                | 2.0 | -31.0                                              | 61.1 | -32.5      | 62.3 | -33.9      | 63.4 | -35.8      | 64.9 |
|                   |      |     | DB                         |      |                    |     |                                                    |      |            |      |            |      |            |      |
|                   |      |     | WB                         |      |                    |     |                                                    |      |            |      |            |      |            |      |

## Monthly Climatic Design Conditions

|                                                                               |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |
|-------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|                                                                               |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                           | DBAvg       | 32.8   | 12.7  | 14.2  | 21.6  | 29.1  | 38.4  | 47.7  | 56.0  | 53.7  | 46.0  | 34.3  | 23.8  | 14.9  |
|                                                                               | DBStd       | 17.22  | 11.02 | 10.87 | 9.10  | 7.39  | 6.25  | 6.26  | 4.24  | 4.46  | 6.13  | 7.88  | 10.33 | 12.60 |
|                                                                               | HDD50       | 6682   | 1157  | 1003  | 882   | 626   | 361   | 114   | 7     | 19    | 145   | 487   | 788   | 1093  |
|                                                                               | HDD65       | 11753  | 1622  | 1423  | 1347  | 1076  | 824   | 520   | 278   | 350   | 569   | 952   | 1237  | 1555  |
|                                                                               | CDD50       | 406    | 0     | 0     | 0     | 0     | 1     | 45    | 194   | 134   | 26    | 0     | 2     | 4     |
|                                                                               | CDD65       | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     |
|                                                                               | CDH74       | 261    | 0     | 0     | 0     | 0     | 0     | 20    | 162   | 71    | 6     | 0     | 0     | 2     |
|                                                                               | CDH80       | 8      | 0     | 0     | 0     | 0     | 0     | 0     | 7     | 1     | 0     | 0     | 0     | 0     |
| Wind                                                                          | WSAvg       | 2.8    | 1.9   | 2.6   | 3.0   | 2.9   | 2.7   | 3.2   | 3.1   | 3.0   | 2.9   | 2.9   | 3.0   | 2.5   |
| Precipitation                                                                 | PrecAvg     | 19.8   | 1.9   | 1.4   | 1.6   | 1.5   | 2.0   | 2.2   | 1.5   | 1.7   | 1.7   | 1.3   | 1.7   | 1.6   |
|                                                                               | PrecMax     | 28.0   | 4.8   | 5.3   | 4.6   | 3.1   | 4.7   | 5.4   | 4.1   | 4.3   | 4.0   | 6.1   | 3.9   | 4.0   |
|                                                                               | PrecMin     | 13.6   | 0.2   | 0.4   | 0.4   | 0.4   | 0.4   | 0.4   | 0.0   | 0.2   | 0.1   | 0.1   | 0.3   | 0.1   |
|                                                                               | PrecStd     | 3.2    | 1.0   | 0.9   | 0.8   | 0.7   | 1.0   | 1.1   | 1.0   | 0.9   | 1.0   | 1.0   | 0.9   | 0.8   |
| Monthly Design<br>Dry Bulb and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 51.8  | 41.4  | 49.2  | 55.2  | 66.2  | 76.8  | 81.0  | 79.3  | 73.9  | 62.3  | 60.6  | 62.9  |
|                                                                               |             | MCWB   | 35.5  | 32.0  | 36.3  | 38.8  | 46.6  | 51.6  | 53.2  | 51.8  | 50.3  | 44.3  | 37.5  | 38.6  |
|                                                                               | 2%          | DB     | 34.4  | 35.7  | 44.3  | 50.7  | 61.1  | 72.6  | 78.2  | 76.0  | 70.2  | 56.9  | 47.4  | 46.3  |
|                                                                               |             | MCWB   | 31.3  | 30.9  | 33.1  | 37.0  | 44.4  | 50.1  | 52.6  | 50.9  | 48.9  | 42.6  | 35.7  | 32.8  |
|                                                                               | 5%          | DB     | 30.1  | 32.4  | 39.4  | 46.4  | 56.6  | 68.9  | 75.6  | 73.0  | 67.0  | 52.3  | 41.8  | 36.1  |
|                                                                               |             | MCWB   | 27.9  | 29.3  | 31.6  | 35.1  | 42.7  | 48.9  | 52.3  | 50.6  | 47.7  | 40.7  | 33.9  | 31.1  |
|                                                                               | 10%         | DB     | 27.2  | 29.9  | 35.6  | 42.3  | 52.3  | 64.8  | 72.7  | 70.3  | 63.2  | 47.7  | 37.4  | 30.5  |
|                                                                               |             | MCWB   | 25.2  | 27.2  | 30.0  | 33.7  | 41.1  | 47.9  | 51.7  | 50.3  | 46.3  | 38.5  | 31.8  | 28.6  |
| Monthly Design<br>Wet Bulb and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 35.2  | 33.6  | 37.1  | 40.9  | 48.7  | 54.8  | 58.7  | 58.2  | 55.0  | 47.3  | 39.9  | 38.5  |
|                                                                               |             | MCDB   | 46.6  | 36.5  | 45.0  | 53.0  | 61.1  | 70.7  | 71.4  | 69.0  | 66.1  | 58.6  | 50.5  | 62.0  |
|                                                                               | 2%          | WB     | 31.4  | 31.6  | 34.6  | 37.8  | 46.2  | 52.0  | 56.5  | 55.6  | 52.0  | 44.2  | 36.9  | 33.4  |
|                                                                               |             | MCDB   | 35.6  | 35.3  | 41.3  | 47.5  | 58.0  | 66.8  | 69.0  | 67.5  | 63.6  | 54.0  | 45.0  | 42.2  |
|                                                                               | 5%          | WB     | 28.2  | 29.7  | 32.5  | 36.1  | 43.9  | 50.2  | 55.1  | 53.9  | 50.0  | 41.8  | 34.5  | 31.5  |
|                                                                               |             | MCDB   | 30.0  | 32.4  | 38.2  | 44.3  | 54.9  | 65.1  | 68.2  | 66.2  | 62.2  | 50.4  | 40.9  | 36.7  |
|                                                                               | 10%         | WB     | 25.5  | 27.3  | 30.7  | 34.4  | 41.7  | 48.6  | 53.7  | 52.3  | 47.9  | 39.3  | 32.6  | 28.7  |
|                                                                               |             | MCDB   | 27.3  | 29.8  | 35.6  | 41.0  | 51.0  | 62.5  | 67.4  | 64.9  | 59.8  | 46.5  | 37.2  | 31.3  |
| Mean Daily<br>Temperature<br>Range                                            | 5% DB       | MDBR   | 19.8  | 22.9  | 24.1  | 22.2  | 21.7  | 27.0  | 31.5  | 31.2  | 28.4  | 21.0  | 19.5  | 18.0  |
|                                                                               |             | MCDBR  | 19.3  | 23.0  | 27.3  | 28.6  | 31.5  | 35.9  | 36.7  | 36.9  | 36.7  | 29.6  | 22.1  | 18.8  |
|                                                                               |             | MCWBR  | 15.1  | 17.8  | 17.9  | 16.2  | 16.3  | 16.4  | 16.3  | 17.4  | 18.7  | 17.0  | 13.0  | 10.8  |
|                                                                               | 5% WB       | MCDBR  | 17.7  | 20.1  | 23.5  | 24.8  | 29.1  | 31.8  | 31.9  | 31.7  | 32.1  | 27.2  | 18.5  | 14.8  |
|                                                                               |             | MCWBR  | 14.0  | 15.7  | 15.7  | 14.5  | 15.8  | 15.5  | 15.7  | 16.7  | 17.6  | 16.6  | 11.7  | 9.6   |
| Clear-Sky<br>Solar<br>Irradiance                                              | taub        | 0.220  | 0.230 | 0.240 | 0.294 | 0.292 | 0.294 | 0.301 | 0.308 | 0.281 | 0.262 | 0.232 | 0.218 |       |
|                                                                               | taud        | 2.369  | 2.330 | 2.353 | 2.297 | 2.505 | 2.572 | 2.584 | 2.529 | 2.616 | 2.616 | 2.522 | 2.396 |       |
|                                                                               | Ebn at Noon | 297    | 311   | 321   | 308   | 310   | 308   | 305   | 300   | 303   | 298   | 291   | 286   |       |
|                                                                               | Edh at Noon | 23     | 30    | 33    | 39    | 33    | 31    | 30    | 31    | 26    | 22    | 20    | 20    |       |
| All-Sky Solar<br>Radiation                                                    | RadAvg      | 529    | 829   | 1219  | 1615  | 1778  | 2103  | 2258  | 1896  | 1466  | 893   | 548   | 410   |       |
|                                                                               | RadStd      | 68     | 92    | 141   | 147   | 207   | 193   | 118   | 105   | 139   | 84    | 65    | 51    |       |

## Historical Trends

|                         | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|-------------------------|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|                         |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|                         | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)   | (l)   | (m)   |
| Station Only            | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| Regional (10 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |

Nomenclature: See separate page